KEVIN BAYU PRADANA

APPLIED BACHELOR OF ROLLING STOCK ENGINEERING

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PROFESSIONAL SUMMARY

Applied Bachelor of Rolling Stock Engineering graduate from Madiun State Polytechnic with a focus on welding technology, metallurgy, manufacturing, and Rollingstock engineering. Experienced in industrial practice at PT INKA (Persero), teaching assistance experience, research, and welding competitions. Certified as a BNSP Welder and Welding Foreman.

EDUCATION

Applied Bachelor of Rolling Stock Engineering **2022 – 2026**
Madiun State Polytechnic
GPA: 3.68 / 4.00

WORK / INTERNSHIP EXPERIENCE

Industrial Internship - Research and Development, PT INKA (Persero) **Jul - Dec 2025**

- Supported technical data analysis, engineering evaluation, and technical documentation in rolling stock manufacturing processes.
- Observed manufacturing and assembly processes and collaborated with engineering teams to support technical problem-solving.

RESEARCH EXPERIENCE

Undergraduate Thesis "Analysis of Laser Power and Wobble Length Variations in Laser Beam Welding on the Mechanical Properties and Metallography of SUS304 Railway Roof Material" **2026**

- Investigated laser power and wobble length effects on lap-shear strength, macrostructure, and microstructure of SUS304 joints.
- Applied experimental design, metallography, mechanical testing, and statistical analysis.

Research Projects - PKM-RE "Alternative Design Optimization of High-Speed Train Mask of Car for Aerodynamic Performance Using CFD Simulation" **2024**

- Contributed as 1 of 4 team members in a research project focused on optimizing high-speed train mask design through Computational Fluid Dynamics (CFD) simulation.
- Participated as Team Member 2 in the development and analysis of alternative train-mask designs for aerodynamic performance.

Research Projects - PKM-AI "Analysis of GMAW Joint Gap Variations on Hardness and Microstructure of Dissimilar S355J2 and SUS 201 Materials" **2024**

- Led a 5-member research team conducting an investigation into the effects of joint-gap variation on hardness and microstructure of dissimilar S355J2 and SUS 201 welds.
- Coordinated 1 PKM-AI research project from team-based research activities focused on welding metallurgy and material characterization.

Research Projects - PKM-AI "Analysis of Mechanical and Physical Characteristics of GMAW-Welded A36 Steel Joints Under Different Cooling Media" **2023**

- Contributed to 1 scientific research project investigating the mechanical and physical characteristics of GMAW-welded A36 steel for railway frame structures.
- Collaborated within a 5-member research team to analyze the influence of cooling media on welded A36 steel joints.

PUBLICATIONS

Optimization of Alternative High-Speed Train Mask-of-Car Design for Aerodynamic Performance at Various Speeds Using Computational Fluid Dynamics (CFD) **2025**
Indonesian Railway Journal, Vol. 9 No. 1 (2025)

- Served as 3rd author in a 7-author research team, published in Jurnal Perkeretaapian Indonesia (Indonesian Railway Journal), Vol. 9 No. 1.
- Analyzed 4 mask-of-car designs at 3 speeds (100, 200, and 300 km/h) using ANSYS Workbench 2023, identifying Design 3 with the lowest drag coefficient of 0.29090604 at 300 km/h.

Applied SMAW Welding Technology to Improve Welding Skills of Students at Miftahul Huda Islamic Boarding School, Doho **2023**
DIKEMAS Journal, Vol. 7 No. 2 (2023)

- Served as 6th author in an 11-member technical training and community-service team, published in JURNAL DIKEMAS, Vol. 7 No. 2 (2023).
- Contributed to training approximately 20 students through SMAW welding theory and practice, including 1F and 2F welding, K3, equipment installation, and product fabrication..

PROJECTS EXPERIENCE

Rail Inspection Development Project – Microprocessors and Microcontrollers 2024

- Developed the **Rail Inspection** system as part of the Microprocessors and Microcontrollers course, focusing on CAD design and component layout.
- Designed the inspection trolley and wheel components using **SolidWorks** and arranged the mechanical components based on the system configuration.

Smart Oven Machine Development – Thermal System & CFD 2024 - 2025

- Designed a smart oven system with controlled temperature using SolidWorks and Autodesk Inventor
- Conducted CFD analysis of airflow and heat distribution using ANSYS Fluent and integrated the heater, blower, and control system.

ORGANIZATION & LEADERSHIP EXPERIENCE

Senior Officer, General Affairs - Student Entrepreneurship Workshop **2024 - 2025**

- Served as 1 of 4 General Affairs positions within the organizational structure, handling administrative and operational affairs for the workshop.
- Contributed to a 1-year organizational term as a Senior Officer supporting the management of the Student Entrepreneurship Workshop.

General Treasurer - Rolling Stock Engineering Student Association (HIMA-TKA) **2024 - 2025**

- Managed organizational financial administration as General Treasurer (Bendahara Umum 1) for 1 term, overseeing budgeting, financial records, and accountability documentation.
- Progressed to General Treasurer after serving as Assistant General Treasurer, completing 2 consecutive treasury roles within HIMA-TKA.

Assistant General Treasurer - Rolling Stock Engineering Student Association (HIMA-TKA) **2023 - 2024**

- Served as Assistant General Treasurer (Bendahara Umum 2) for 1 organizational term, supporting financial administration and treasury operations.
- Built 2 years of consecutive organizational finance experience, progressing from Assistant General Treasurer to General Treasurer.

COMMUNITY SERVICE

Volunteer - Student Organization Capacity Building Program (PPK Ormawa) – Tambakmas Village **2025**

- Served as 1 volunteer in the Student Organization Capacity Building Program (PPK Ormawa) implemented in Tambakmas Village.
- Contributed to 1 community empowerment program conducted over a 4-month period, from July to October 2025.

Competitive Internal Service Program – GMAW Gasless Welding Product Repair **2024**

- Served as Member 4 in an 8-member community service team conducting product repair activities using GMAW Gasless welding.
- Contributed to 1 community service project conducted over approximately 7 months, focused on improving welding product quality and practical welding skills.

Competitive Internal Service Program – Applied SMAW Welding **2023**

- Served as Member 4 in a 9-member community service team implementing applied SMAW welding activities.
- Contributed to 1 community service project conducted over approximately 6 months, focused on applied SMAW welding training and practical skill development.

Student Community Empowerment Program (P2MD) – Batch 2, Bader Village **2023**

- Contributed as 1 of 10 team members from HIMA-TKA, Madiun State Polytechnic, in implementing a community empowerment program in Bader Village, Madiun Regency.
- Participated in 1 P2MD-funded program focused on applying appropriate technology and business digitalization concepts to transform organic waste into fertilizer.

Student Community Empowerment Program (P2MD) – Batch 1, Bedingin Village **2023**

- Contributed as 1 of 10 team members from HIMA-TKA, Madiun State Polytechnic, in implementing a community empowerment program in Bedingin Village, Ponorogo Regency.
- Participated in 1 P2MD-funded program focused on applying appropriate technology and business digitalization concepts to convert organic waste into fertilizer.

TEACHING ASSISTANT EXPERIENCE

Teaching Assistant – Engineering Drawing **2025**

- Served as a teaching assistant for 1 academic semester, covering 16 practicum sessions/modules from basic drawing standards through AutoCAD Inventor projects.
- Reviewed and evaluated 3 AutoCAD Inventor projects and supervised practical assignments covering geometric construction, projections, section views, tolerances, surface configuration, threaded components, bolts, springs, and welded joints.

Teaching Assistant – Occupational Health and Safety (K3) **2025**

- Served as a teaching assistant for 1 academic semester, assisting the implementation and supervision of K3 standards in classroom and laboratory environments.
- Facilitated 4 core K3 teaching activities, including course introduction, learning evaluation, case-study discussions, and final course evaluation.

Manufacturing Workshop & Laboratory Assistant **2025**

- Served as a workshop and laboratory assistant across 5 academic semesters, supporting laboratory safety, equipment maintenance, SOP compliance, troubleshooting, and research activities.
- Conducted 8 core laboratory responsibilities, including K3 monitoring, risk assessment, equipment maintenance, troubleshooting, equipment demonstrations, SOP supervision, housekeeping, and material readiness.

CERTIFICATIONS & TRAINING

Certified Welder – Level 3 BNSP Certificate of Competency **2026**

- Obtained Level 3 Welder competency certification under the Mechanical Engineering classification and Operator qualification.
- Earned a professional competency certificate valid for 5 years, issued through the BNSP certification system.

Certified Welding Foreman BNSP Certificate of Competency **2026**

- Achieved “Competent” status in the Welding Foreman competency assessment conducted on 20 July 2026.
- Successfully qualified as 1 of 15 candidates assessed under the Welding Foreman scheme.

Basic Railway Welding Training – Indonesian Railway Polytechnic **2026**

- Completed 24 instructional hours of basic railway welding training and was declared to have passed.
- Completed 4 training modules, covering welding fundamentals, welding techniques and positions, welding K3, and field work practice.

Basic Excavator Operation Training – Indonesian Railway Polytechnic **2026**

- Completed 24 instructional hours of basic excavator operation training and was declared to have passed.
- Completed 4 training modules covering excavator specifications, heavy-equipment K3, excavator operation techniques, and excavator maintenance and inspection.

ACHIEVEMENTS

3rd Place – National Welding Competition 2026, Expert Level, GMAW Plate 3G. **2026**

- Achieved 3rd Place in the Expert Level – GMAW Plate 3G category at the National Welding Competition 2026.
- Competed at Politeknik Perkapalan Negeri Surabaya in a national-level welding competition involving the GMAW Plate 3G welding category.

TECHNICAL SKILLS

Welding & Material Testing

- Welding Technology
- GMAW
- SMAW
- Laser Beam Welding
- Welding Inspection
- Non-Destructive Testing (NDT)
- Welding Safety / K3
- Metallography
- Macro Examination
- Microstructure Analysis
- Tensile Testing
- Vickers Hardness Testing

Manufacturing & Engineering

- Manufacturing Process
- Production Process
- Fabrication

- Quality Control
- Engineering Documentation
- Technical Drawing

Engineering Software

- AutoCAD
- Autodesk Inventor
- SolidWorks
- ANSYS Workbench
- OriginPro
- SPSS
- Microsoft Excel
- Microsoft Word

SOFT SKILLS

- Technical Problem Solving
- Critical Thinking
- Teamwork & Collaboration
- Leadership
- Communication
- Time Management
- Adaptability
- Safety Awareness
- Decision Making

LANGUAGES PROFICIENCY

English Proficiency Test (EPT) **2026**

- Achieved an EPT score of 537, demonstrating standardized English language proficiency.
- Earned a standardized English proficiency score for use as an academic and professional qualification.

English Proficiency – TOEFL ITP **2026**

- Achieved a TOEFL ITP Level 1 score of 450, demonstrating English proficiency across Listening Comprehension, Structure & Written Expression, and Reading Comprehension.
- Recorded section scores of 54 Listening, 35 Structure & Written Expression, and 46 Reading Comprehension.

LANGUAGE

Indonesian - Native

English - EPT 537 | TOEFL ITP Level 1: 450